

LakbayLokal

Go Beyond, Go Lokal.

Offline-First Tourist Trail and Destination Information Mobile System for Davao del Norte

Project Overview

LakbayLokal is a mobile tourism application designed for travelers exploring Davao del Norte, especially destinations where internet connectivity may become weak or unavailable. Instead of depending on live online services throughout the trip, the application lets users download destination-specific tourism packs in advance and continue using essential travel information offline.

Focus	Offline-first local tourism support
Primary Area	Davao del Norte
Main Framework	React Native with TypeScript
Core Principle	Download before travel; continue exploring beyond the signal
Cost Goal	Core prototype uses free/open-source technologies and no required paid API

1. Problem

Tourists may lose access to useful travel information once they enter areas with unstable mobile data or weak signal coverage. This can make it difficult to locate attractions, follow trails, review transport instructions, or access basic safety information at the exact moment those details are needed.

- Online maps and tourism pages depend heavily on connectivity for their full functionality.
- Visitors unfamiliar with local destinations may not know the correct trail, route, or next stop when signal disappears.
- Tourism information is often scattered across websites, social pages, and map listings instead of being prepared as one offline package.
- General-purpose map applications are not designed specifically around curated Davao del Norte tourism packs.

2. Solution

LakbayLokal provides **downloadable destination packs**. Before traveling, the tourist downloads the pack for a selected area. The app stores the map content, tourist trail data, destination details, transport notes, images, and safety information on the phone. During the trip, the phone's GPS can still determine the user's position while the downloaded content remains available without internet.

- Download a destination pack while connected.
- Store maps, points of interest, trail data, and tourism information locally.
- Use GPS positioning and predefined tourist trails while offline.
- Access local destination, transport, and safety information even without mobile data.
- Synchronize or update packs again when internet becomes available.

3. Beneficiaries

- **Tourists and visitors** - gain a more reliable way to explore local destinations when connectivity is uncertain.
- **Local tourism communities** - gain a dedicated channel for curated local attraction, cultural, and travel information.
- **Micro and small local businesses** - can benefit from tourists remaining informed about nearby services and establishments.
- **Tourism organizers and destination managers** - may optionally contribute verified destination information without being required for the app to function.

4. Core Features

Feature	Purpose
Destination Pack Download	Downloads all required tourism content for a selected destination before the trip.
Offline Tourist Map	Shows attractions, facilities, and important locations using locally stored map resources.
Offline Tourist Trails	Displays predefined walking/travel trails using stored GeoJSON and the device's GPS.
Offline Destination Guide	Provides descriptions, reminders, images, operating information, and local context without internet.
Offline Safety and Transport Notes	Stores essential travel notes, local transport instructions, and emergency information for quick access.

5. Technology Used

The architecture is designed around free and open-source tools. React Native remains the main application framework. The other technologies support offline maps, device location, local storage, destination-pack packaging, and development workflow.

Technology	Role	Cost
React Native + TypeScript	Main mobile application framework and programming language setup.	Free
Expo	Development workflow, device APIs, local testing, and optional build tooling.	Free for core use
MapLibre React Native	Renders map content and supports offline map resources.	Free / open source
OpenStreetMap Data	Base geographic data used when preparing map content.	Free with attribution
Device GPS / Expo Location	Determines user location and updates position while offline.	Free; uses phone hardware
SQLite	Stores downloaded packs, destinations, points of interest, trail metadata, and offline records.	Free
Local File Storage	Stores images, JSON, GeoJSON, and downloaded pack assets directly on the device.	Free
GeoJSON	Represents tourist trails, route lines, points, and geographic boundaries.	Free / open format
JSON	Structures destination-pack information such as descriptions, transport notes, and pack metadata.	Free / open format
Expo Router / React Navigation	Manages application screens and navigation flow.	Free
Node.js + Express	Optional backend for publishing and updating tourism packs in a later version.	Free software; hosting optional
Git + GitHub	Version control and team collaboration during development.	Free plan available

Technology Cost Rule for Version 1

Use	Avoid / Not Required
React Native, Expo, MapLibre, OpenStreetMap-derived offline data, GPS, SQLite, JSON, GeoJSON, local files	Google Maps paid API, paid AI services, paid OCR, Firebase dependency, paid cloud server, special hardware, IoT devices

Important map note: OpenStreetMap data is open, but the public OpenStreetMap tile server should not be used for bulk offline downloading. LakbayLokal should prepare or package its own limited offline map resources for the selected prototype destinations, then render those resources through MapLibre.

6. Proposed Offline-First Architecture

When Internet Is Available	When Internet Is Unavailable
User selects a destination pack and downloads map resources, destination information, trail GeoJSON, images, transport notes, and safety content.	MapLibre loads local map resources; GPS provides the user's current position; SQLite and local files provide all destination and trail information.
Optional backend can later publish versioned pack updates.	No live backend, paid API, or cloud request is required for the core downloaded pack to function.

ONLINE Download destination pack	->	**LOCAL DEVICE** Map + SQLite + JSON + GeoJSON + Images	->	**OFFLINE TRIP** GPS + local content continue working

7. Theme and Logo Direction

The approved theme is **Modern Local Adventure**. The visual identity should feel nature-inspired, approachable, and distinct from mainstream map applications.

Element	Approved Direction
Primary Colors	Forest Green + Warm Orange + Off-white
Brand Mood	Local, adventurous, friendly, dependable, and modern
Logo Concept	Location Pin + Mountain with a winding trail and subtle nature elements
Tagline	Go Beyond, Go Lokal.



The location pin represents destination and place. The mountain and trail communicate local exploration and outdoor tourism. Forest green reflects nature and reliability, warm orange adds energy and adventure, while off-white keeps the visual identity clean and welcoming.

8. Difference Between LakbayLokal and Google Maps

LakbayLokal should not be presented as a replacement for Google Maps. Google Maps remains a stronger general-purpose navigation platform. LakbayLokal focuses on a narrower problem: **curated destination-specific tourism information that remains useful when connectivity becomes unreliable**.

Aspect	Google Maps	LakbayLokal
Primary Purpose	General mapping and navigation	Offline-first tourism companion for Davao del Norte
Offline Map Areas	Available through downloaded areas	Included as part of a complete destination pack
Offline Tourist Trails	Not designed around curated local trail packs	Predefined local tourist trails stored as offline geographic data
Offline Local Transport Instructions	Not focused on destination-pack transport notes	Curated instructions can be stored with each destination pack
Tourism Content	Broad/global place information	Curated Davao del Norte attraction and local travel content
Safety Information	Not organized as destination-specific offline tourism packs	Can be packaged with emergency contacts, reminders, and local safety notes
Local Identity	Global platform	Built specifically around local destinations and local context
Core Strength	Comprehensive online/general navigation ecosystem	Prepared offline tourism experience for selected local destinations

9. Recommended Prototype Scope

- Start with selected Davao del Norte destinations rather than attempting to cover the entire province immediately.
- Prioritize Samal and a small number of other destinations where offline access can be demonstrated clearly.
- Keep Version 1 limited to destination packs, offline map/trail use, GPS positioning, destination information, and safety/transport notes.
- Do not add booking, payments, chat, social media, AI recommendation, or complicated user-account features unless later required.

LakbayLokal - *Go Beyond, Go Lokal.*